

We shall deal with positive form as linear operator at first, and we will generalize.

Def. A linear operator $T: V \rightarrow V$ on a fin. dim inn. prod space V is called positive if:

T is Hermitian and $\langle T(\alpha), \alpha \rangle \geq 0$ for all $\alpha \neq 0$

Prop. Let T, U be Hermitian on fin. dim inn. prod space V .

And, $A = [T]_{\beta}$, where β is any orthonormal basis for V .

1). T is positive if and only if all eigenvalues of T are positive.

2). T is positive if and only if $\sum_{i=1}^n A_{ii} a_i \bar{a}_i \geq 0$ for all nonzero tuple $(a_1, \dots, a_n)$.

3). T is positive if and only if

$$A = B^* B \text{ for some } B.$$

Proof. Denote $\beta = \{\alpha_1, \dots, \alpha_n\}$, Let $B = [I]_{\beta}^{\{\alpha_1, \dots, \alpha_n\}} = (\alpha_1 \ \alpha_2 \ \dots \ \alpha_n)$.

And, let $\beta' = \{\beta_1, \dots, \beta_n\}$ be an orthonormal basis for V consisting eigenvectors of T .

1). If T is positive, then $\langle T(\beta_i), \beta_i \rangle = \langle C_i \beta_i, \beta_i \rangle = C_i \geq 0$, where C_i is eigenvalue.

Conversely, given $X \in V$, $X = x_1 \beta_1 + \dots + x_n \beta_n = \langle X, \beta_1 \rangle \cdot \beta_1 + \dots + \langle X, \beta_n \rangle \cdot \beta_n$, so

$$\begin{aligned} \langle T(X), X \rangle &= \langle \langle X, \beta_1 \rangle \cdot C_1 \beta_1 + \dots + \langle X, \beta_n \rangle \cdot C_n \beta_n, \langle X, \beta_1 \rangle \cdot \beta_1 + \dots + \langle X, \beta_n \rangle \cdot \beta_n \rangle \\ &= \sum_{i=1}^n C_i \cdot |\langle X, \beta_i \rangle|^2 \geq 0 \text{ if } X \neq 0 \text{ and } C_i \geq 0. \end{aligned}$$

2). Given $X \in V$ st $[X]_{\beta} = (a_1, \dots, a_n)$,

$$\langle T(X), X \rangle = \left\langle \sum_{j=1}^n a_j T(\alpha_j), \sum_{i=1}^n a_i \alpha_i \right\rangle = \sum_{j=1}^n \sum_{i=1}^n \langle T(\alpha_j), \alpha_i \rangle \cdot a_j \cdot \bar{a}_i, \text{ so proved.}$$

(+Recall: $A_{ij} = \langle T(\alpha_j), \alpha_i \rangle$).

3). If $A = B^* B$, then $\langle B^* B x, x \rangle = \langle Bx, Bx \rangle = \|Bx\|^2 \geq 0$. Conversely,

since $[T]_{\beta} = D$ is diagonal with positive values, where

$$D = \begin{pmatrix} C_1 & & 0 \\ & C_2 & \\ 0 & & C_n \end{pmatrix} \text{ } C_i \text{'s are eigenvalues of } T$$

so we can define $\sqrt{D} = \begin{pmatrix} \sqrt{C_1} & & 0 \\ & \sqrt{C_2} & \\ 0 & & \sqrt{C_n} \end{pmatrix}$. Now, for $P = [I]_{\beta}^{\beta}$,

$$A = P^{-1} D P = P^{-1} \cdot \sqrt{D} \cdot \sqrt{D} \cdot P = (\sqrt{D} P)^* (\sqrt{D} P), \text{ since } P \text{ is unitary, } \sqrt{D} \text{ is symmetric.}$$

And, there are more interesting problems,

4). If T and U are positive s.t. $T^2 = U^2$, then $T = U$.

5). If T and U are positive s.t. $TU = UT$, then TU is positive.

Proof. Let $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis for V consisting of eigenvectors of T . Let $C_1, \dots, C_n$ be eigenvalues corresponding to $\alpha_1, \dots, \alpha_n$, respectively. Setting for each $1 \leq i \leq n$,

$$T^2(\alpha_i) = C_i^2 \cdot \alpha_i = U^2(\alpha_i).$$

$$\text{And, } C_i^2 \alpha_i = U^2(\alpha_i) \Leftrightarrow (U^2 - C_i I)(\alpha_i) = 0.$$

$$\Leftrightarrow (U + C_i I) \circ (U - C_i I)(\alpha_i) = 0.$$

If $V = (U - C_i I)(\alpha_i) \neq 0$, then

$$U(V) = -C_i V$$

So V is an eigenvector of U with negative eigenvalue.

It is contradiction to U is positive, so $U(\alpha_i) = C_i \alpha_i$ for all $1 \leq i \leq n$.

The linear map is determined by maps basis to some vectors, so $T = U$ (4)

Since T and U commute and both are Hermitian,

$$(TU)^* = U^* T^* = UT = TU$$

So TU is Hermitian. For the above $\mathcal{B}$, define a linear map as

$$\sqrt{T}(\alpha_i) = \sqrt{C_i} \cdot \alpha_i.$$

Then, for each $1 \leq i \leq n$,

$$U\sqrt{T}(\alpha_i) = U(\sqrt{C_i} \alpha_i) = \frac{1}{\sqrt{C_i}} \cdot U(C_i \alpha_i) = \frac{1}{\sqrt{C_i}} \cdot U(T(\alpha_i)) = \frac{1}{\sqrt{C_i}} \cdot TU(\alpha_i)$$



Def. A form f on an inn. prod space V is called non-negative if:

f is Hermitian and $f(\alpha, \alpha) \geq 0$ for all $\alpha \in V$.

and, if $f(\alpha, \alpha) > 0$, $\alpha \neq 0$, it is called positive,

Exmple. Let f be a form on fin. dim inn. prod space V , let $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ be an ordered basis.

Let A be a matrix of f in $\mathcal{B}$. That is, $A_{jk} = f(\alpha_k, \alpha_j)$

If $\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n$, then

$$f(\alpha, \alpha) = f\left(\sum_j x_j \alpha_j, \sum_k x_k \alpha_k\right) = \sum_j \sum_k x_j \overline{x_k} \cdot f(\alpha_j, \alpha_k) = \sum_j \sum_k A_{kj} \cdot x_j \cdot \overline{x_k}.$$

So, f is non-negative if and only if $A = A^*$ and $\sum_j \sum_k A_{kj} x_j \overline{x_k} \geq 0$, for all x_j, x_k

Equivalently, f is non-negative if and only if

$$\mathcal{L}: F^n \times F^n \rightarrow F: (X, Y) \mapsto Y^* A X$$

is non-negative form, $F = \mathbb{C}$ or $\mathbb{R}$,

Thm. Let $F = \mathbb{C}$ or $\mathbb{R}$, and $A \in M_{\text{sym}}(F)$.

A function $\mathcal{L}: F^n \times F^n \rightarrow F: (X, Y) \mapsto Y^* A X$

is a positive form if and only if $A = P^* P$ for some invertible $P \in M_{\text{inn}}(F)$.

Proof Suppose that $\mathcal{L}$ is positive form. Then, it is an inner product on F^n .

So there is an orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$ for F^n , satisfying

$$\mathcal{L}(\alpha_i, \alpha_j) = \alpha_i^* A \alpha_j = \delta_{i,j}.$$

so, take $P = (\alpha_1 \dots \alpha_n)^{-1}$, then $P A P^* = I$, or $A = P^* P$.

Conversely, if $A = P^* P$ for some invertible P , then $A^* = (P^* P)^* = P^* P = A$ and

$$X^* A X = X^* P^* P X = (P X)^* \cdot (P X) \geq 0 \quad \text{if } X \neq 0.$$

So proved.

Def. Let $A \in M_{n \times n}(F)$. The principal minors of A are:

$$\Delta_k(A) \stackrel{\text{def}}{=} \det \begin{bmatrix} A_{11} & \dots & A_{1k} \\ \vdots & \ddots & \vdots \\ A_{k1} & \dots & A_{kk} \end{bmatrix} \quad 1 \leq k \leq n$$

Lemma. Let $A \in M_{n \times n}(F)$ be an invertible. TFAE:

- There is a uni-upper triangle matrix P such that AP is lower triangular.
- The principal minors of A are all different from 0.

Proof. Note: B is upper triangle iff $B_{jk} = 0$ if $j > k$.

Suppose that a) holds.

Denote $A_1, \dots, A_n$ are columns of A , and $B_1, \dots, B_n$ are column of $B = AP$.

Then, $B_1 = A_1$ by P is uni-upper, and

$$B_r = A_r + \sum_{j=1}^{r-1} p_{jr} \cdot A_j$$

(observe that

$$\begin{aligned} (B_1 \dots B_n) &= (A_1 \dots A_n) \cdot \left(I + \begin{pmatrix} 0 & & p_{1k} \\ & \ddots & \\ 0 & & 0 \end{pmatrix} \right) \quad (p_{jk} = 0 \text{ if } j \geq k) \\ &= (A_1 \dots A_n) + \underbrace{\left(\sum_{j=1}^{r-1} p_{jr} \cdot A_j \right)}_{r\text{th column}} \end{aligned}$$

Therefore, for each $1 \leq k \leq n$,

$$\begin{bmatrix} B_{11} & \dots & B_{1k} \\ \vdots & \ddots & \vdots \\ B_{k1} & \dots & B_{kk} \end{bmatrix} \text{ is obtained by adding to the } r\text{th column of}$$

$$\begin{bmatrix} A_{11} & \dots & A_{1k} \\ \vdots & \ddots & \vdots \\ A_{k1} & \dots & A_{kk} \end{bmatrix} \text{ as a linear combination of its other columns.}$$

This operation cannot change the determinant, so, $\Delta_k(A) = \Delta_k(B)$.

And, since A, P are invertible, so is B , thus $\Delta_n(B) = \det B \neq 0$.

Now, $\Delta_k(B) = B_{11} \dots B_{kk} \neq 0$, since it is triangular.